TIC 65257021

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period       4.790576 d
  Epoch (t0)   1492.8520 BTJD
  Duration     0.0840 d
  Depth        4.348 %  (43,484 ppm)
  SNR          118.7

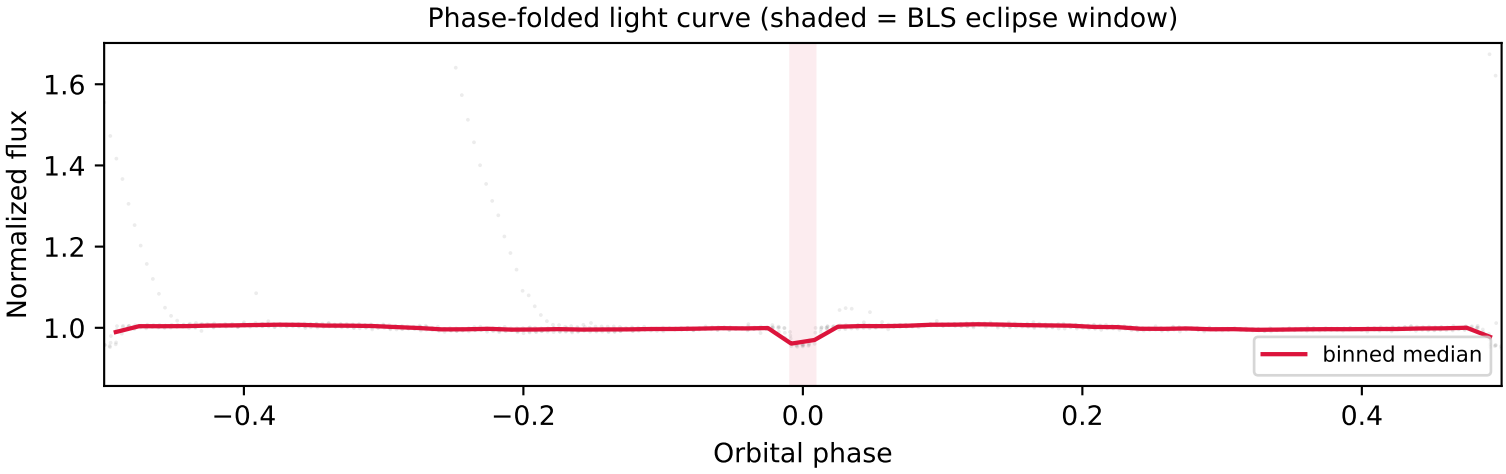
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     4  (gate >= 3)
  Robust depth ratio    0.96  (gate >= 0.50)
  Robust depth sigma    23.1  (gate >= 7.0)
  Odd/even z-score      0.05  (consistent)
  Secondary eclipse     detected, phase 0.49, 5.7 sigma
```

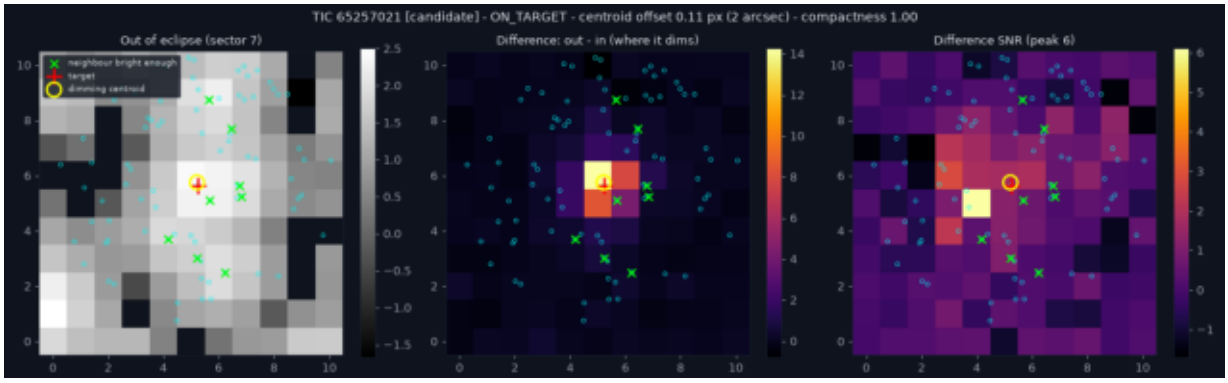
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 7    light-curve dip in this sector: 20.2 sigma    t0 refined by -0.0025 phase
  Difference-image peak SNR   6.10   (gate >= 5.0)      Compactness   1.00   (gate >= 0.50)
  Dimming centroid offset     0.11 px (2 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 65257017
    0.81 px away, dTmag +1.34 - farther from the dimming than the target
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No caveats - clean detection on every check.